

Small Data Insights for Groundwater Management

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Data scarcity poses significant challenges in environmental fields, including agricultural systems,¹ ecosystem management,² and water resource assessment,³ where incomplete or fragmented data sets often hinder accurate analysis and decision-making. These challenges are particularly pronounced in groundwater systems, where the concealed nature of subsurface environments, logistical difficulties, and high monitoring costs severely limit data collection. Furthermore, the spatial heterogeneity of geological formations and the fragmented nature of available data sets amplify uncertainties in modeling groundwater flow and contaminant transport. Such complexities necessitate the development of a small data approach that can extract meaningful insights from limited data sets, enabling timely, cost-effective decision-making in groundwater management and enhancing the efficiency of environmental remediation efforts under data-constrained conditions.

ORIGINS AND ISSUES OF SMALL DATA

The hidden characteristics of subsurface systems create significant challenges for directly observing soil and ground-

water contamination, particularly in complex underground media.⁴ Monitoring techniques, such as borehole sampling and intermittent water quality testing, provide localized data but are often fragmented by spatiotemporal discontinuities⁵ (Figure 1). This results in substantial data gaps that increase uncertainties in contaminant migration studies in heterogeneous environments with heterogeneity in geological properties.⁶ Additionally, the confined nature of groundwater flow often delays contaminant detection, complicating efforts to accurately trace source information.⁷ These challenges along with the high costs and logistical difficulties of continuous monitoring frequently result in small data scenarios. Furthermore, gathering

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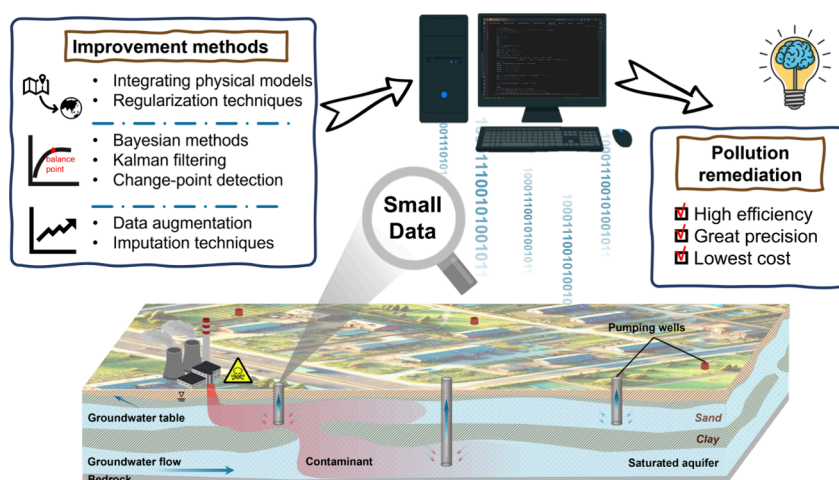


Figure 1. Small data generation, enhancement methods, and data analysis within groundwater systems are essential for improving remediation outcomes at contaminated sites. Constrained by high sampling costs and logistical challenges, limited monitoring data collected from boreholes can be leveraged for in-depth analysis at a site with contaminated groundwater. Optimized methods for maximizing the utility of small data not only enhance the evaluation of subsequent remediation efforts but also support the development of efficient, high-precision, and cost-effective remediation strategies.

large volumes of data quickly is rarely feasible in practical applications, especially under time constraints.⁸ Despite their limitations, small data sets often serve as the only available information during critical moments.⁹ Consequently, their importance increases as they offer timely and actionable insights that align with the immediate needs of real-world engineering and environmental management. Therefore, effectively leveraging these limited data sets is essential for decision-makers to respond efficiently, optimizing resource allocation and environmental impact.

■ NECESSITY OF OPTIMIZING SMALL DATA UTILIZATION

Researchers have explored the impact of varying the number of pumping tests and monitoring wells to understand how additional data points influence model accuracy in the field of groundwater system characterization. For instance, increasing the number of pumping tests from four to nine and the number of monitoring wells from 49 to 158 yields a <10% improvement in predictive accuracy,^{10–12} highlighting the diminishing returns of additional data volume. This phenomenon of diminishing returns suggests that after a balance point,⁹ accumulating more data adds limited value, especially when weighed against the associated costs in time, equipment, and computational complexity.⁸ Additionally, the expanding data volume can inadvertently introduce redundancy, saturating information to the point where the marginal utility of each new data point decreases.¹³ Such redundancy not only increases the time and resources required for model training but also escalates the risk of overfitting, ultimately reducing generalizability and prediction reliability in real-world applications.

Researchers have applied Bayesian–MCMC (Markov chain Monte Carlo) methods to pinpoint critical balance points in data accumulation,⁹ ensuring that model accuracy is maximized without incurring disproportionate costs or resource demands. Identifying this optimal data threshold becomes essential for guiding practical engineering decisions, particularly in resource-constrained scenarios in which efficiency in data use is paramount. Consequently, small data approaches not only enhance decision accuracy but also significantly reduce costs and improve decision-making efficiency. While methods that rely on

large-scale data collection require substantial time and resources for decision-making, a small data approach can extract valuable insights from limited data, enabling quicker responses and more efficient decisions. This is particularly important in resource-constrained environments, where small data approaches help decision-makers strike a better balance between data acquisition and cost investment. By optimizing resource allocation, small data approaches ensure more forward-thinking and reliable decisions in groundwater management. Ultimately, the application of small data approaches improves decision efficiency while maintaining accuracy, providing robust support for sustainable development.

■ LESSONS FROM SMALL DATA APPLICATIONS ACROSS FIELDS

Recognizing the importance of small data research, existing studies show that small data-based prediction techniques can deliver accurate contaminant migration models even without extensive data sets, offering reliable guidance for remediation strategies.¹⁴ For instance, in molecular science, in which data acquisition can be highly constrained, machine learning models like random forests and support vector machines have been applied to predict drug–target interactions and drug toxicity with promising results.^{15–17} Similarly, researchers have addressed small data challenges by utilizing data augmentation and transfer learning, combining convolutional neural networks with molecular image data to enhance quantitative structure–activity relationship modeling, enabling accurate predictions of molecular quantum structures and activity relationships in the field of chemistry.^{18,19} Ecology has similarly benefited from small sample learning, where it has played a crucial role in understanding habitat requirements and population dynamics of rare species, supporting biodiversity conservation efforts.^{20,21} Meanwhile, the shift toward personalized production and small batch manufacturing has highlighted the limitations of traditional big data-based predictive models, which are often prone to overfitting in quality management. To counter this, researchers have incorporated data augmentation techniques, such as recurrent variational autoencoders and conditional generative adversarial networks, alongside optimization algorithms like proximal policy optimization and support vector regression.

This approach has improved product quality predictions in small sample environments.^{22,23} Additionally, in environmental science and hydrological modeling, innovative methods combining long short-term memory networks with prototypical networks have been successfully employed to address runoff prediction challenges in data-scarce river basins, effectively overcoming the limitations posed by sparse hydrological data.^{24,25} With respect to hydrogeology, timely optimization of resource utilization and minimization of response times can enhance the understanding of contaminant migration in groundwater systems. Approaches that focus on extracting insights from limited data sets have significant potential, not only in addressing challenges in predictive medicine, ecology, and manufacturing but also in advancing the analysis of data-limited systems in fields like climate modeling, urban planning, and biodiversity monitoring.

■ SMALL DATA OPPORTUNITIES IN GROUNDWATER

Building on the achievements of small data applications across disciplines, promising and underutilized opportunities exist within hydrogeology, particularly in complex and concealed groundwater systems where traditional large data approaches often fall short. Small sample learning has demonstrated its value in managing data scarcity in fields such as molecular science, where predictive accuracy is achieved through limited data sets. However, similar approaches are not yet widely adopted in hydrogeological studies, where modeling often depends on well-established, data-intensive models like MODFLOW and MT3DMS or on machine learning surrogates trained with extensive data sets.²⁶ These models can become unreliable in data-limited situations, highlighting a gap that small data techniques can address.

Due to the concealment and heterogeneity of aquifers, as well as logistical constraints and high costs associated with data collection, groundwater systems present an effective test case for small data methods.^{27,28} These factors hinder large data set acquisition, thereby increasing uncertainty in contaminant transport modeling. Furthermore, traditional hydrogeological surveys face challenges from geological heterogeneity and limited monitoring data.^{29,30} Common interpolation methods, such as kriging and inverse distance weighting, assume data stationarity, which limits their ability to address data complexity and nonstationarity.²⁷ While expanding the monitoring network can improve data availability, the high costs and logistical challenges restrict the potential for accuracy improvements in data-scarce situations.³¹

Given these limitations, hydraulic tomography (HT) differs from interpolation methods by directly capturing the heterogeneity and connectivity of permeable zones, enhancing parameter estimation and contaminant transport predictions under data-scarce conditions.^{32–35} Despite these advantages, residual uncertainties remain a major challenge under small data conditions. To address these uncertainties, advanced optimization methods like genetic algorithms, particle swarm optimization, and Bayesian optimization are increasingly used to improve contaminant distribution characterization.²⁷ Among these, Bayesian inversion excels at refining posterior distributions and mitigating uncertainties, particularly in small data scenarios.^{30,31} Therefore, exploring the combination of these technologies can characterize both the heterogeneity of the aquifer and the distribution of contaminants, while quantifying the associated uncertainties. This approach could establish a promising framework that plays a significant role in groundwater

management, where small data phenomena are common. Building on these advancements, small data approaches have proven to be valuable in not only groundwater modeling but also other fields of groundwater research. Data scarcity often arises in environmental monitoring due to limited sampling points, low monitoring frequency, and the infrequency of contamination events, which impacts the timely and accurate identification of sources and contaminant distribution predictions.³⁶ To address this issue, small data methods such as transfer learning, generative adversarial networks (GANs), and data augmentation can extract knowledge or generate additional data from limited data sets,^{37,38} filling data gaps and thereby enhancing model generalization and predictive accuracy (Figure 1).

■ CHALLENGES AND PROSPECTS

Achieving model accuracy and generalizability is fraught with several distinct challenges in small data scenarios within concealed groundwater systems. First, the heterogeneity of aquifer parameters and the spatial variability of hydrogeological properties hinder model generalization, causing overfitting and reduced accuracy in broader applications.³⁹ Second, as the volume of data increases, the marginal benefit of additional data diminishes, making it difficult to identify the point at which further data introduce redundancy and additional costs without significantly improving model performance.⁹ Lastly, determining the optimal volume of data for groundwater management is crucial to balance model accuracy and resource efficiency, enabling cost-effective practices without compromising predictive capabilities.

Addressing these challenges requires strategies to enhance model robustness, including integrating physical models with regularization techniques and data augmentation to reduce overfitting and improve generalizability.^{14,40} These methods allow models to adapt more effectively to diverse hydrogeological conditions and mitigate overfitting risks. Advanced Bayesian and filtering methods optimize data use by dynamically estimating parameters and quantifying uncertainty,^{9,41} allowing for efficient monitoring with limited data sets and identifying when further data collection becomes redundant. In combination, expanding data sets through augmentation and imputation,¹⁵ alongside ensemble learning techniques,⁴² further enhances model stability.⁴³ In addition, semisupervised techniques enhance generalization by leveraging unlabeled data.⁴⁴

While the existing methods offer advantages in addressing the challenges posed by small data, they still have inherent limitations. Regularization techniques designed to mitigate overfitting in data-scarce scenarios may not effectively address data gaps in regions with significant heterogeneity.¹⁴ The extreme variability in hydraulic properties in such areas means that regularization cannot fully capture local differences, and data augmentation may not accurately replicate this complexity, potentially introducing artificial patterns that fail to reflect the true characteristics of the aquifer.¹⁵ Similarly, Bayesian methods are valuable for uncertainty quantification and are highly dependent on the quality of prior knowledge.⁴⁵ In groundwater studies with sparse or imprecise prior information, this dependency can bias posterior distributions, compromising parameter estimates like hydraulic conductivity. When prior data are limited or unreliable, Bayesian methods may yield misleading results, complicating decision-making in groundwater management.⁴⁶ Advances in small data methodologies are transforming groundwater management by enhancing the

understanding of complex systems, enabling precise environmental control, and fostering sustainable practices. These innovations address data-limited challenges, promote resilient ecosystems, and support informed, long-term resource conservation policies.

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Author Contributions

Z.Z. and Y.W. led the conceptualization, writing, and figure drafting. Y.C., Y.C., and J.Z. assisted with writing and figures.

H.L. and Y.W. conceived the ideas and designed the framework.

T.-C.J.Y. and Y.L. helped to improve the writing of the paper. All authors approved the final form for publication.

Notes

The authors declare no competing financial interest.

Biography



Dr. Yaqiang Wei is currently an Associate Professor at Shanghai University. He completed his postdoctoral research at Shanghai Jiao Tong University and obtained his Ph.D. from the University of Chinese Academy of Sciences in 2017, with a joint training program at the University of Arizona, United States. His primary research focus is on the migration, transformation, and fate of contaminants in soil and groundwater. He has led several research projects, including a project of key fund and a Youth Fund from the National Natural Science Foundation, and two subprojects under the National Key Research and Development Program for Soil Pollution Causes and Control Technologies. He also serves as a young editorial board member for journals such as *Eco-Environment & Health* and *Agriculture Communications*.

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